

Applied Cloud Computing

Assignment 3: Amazon CLI, Webhosting in S3, Lambda

Project Deliverable

1. Document the creation of AWS bucket and object using aws CLI that includes: (15 pts)

1.1 Screenshots(use '**partial screen capture**' – Mac is "Command+Shift+4", Windows is "Windows + Shift + S") of each execution of the aws CLI commands and outputs at step 5 (10 pts)

- `// create a bucket`

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % aws s3 mb s3://shwetaleegadekar-cloudcompute-oct2025
make_bucket: shwetaleegadekar-cloudcompute-oct2025
```
- `// list bucket(s) created`

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % aws s3 ls
2025-10-13 14:31:35 shwetaleegadekar-cloudcompute-oct2025
```
- `// copy a local file into s3 and create an object in your bucket`

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % aws s3 cp hw3-testfile.txt s3://shwetaleegadekar-cloudcompute-oct2025/
upload: ./hw3-testfile.txt to s3://shwetaleegadekar-cloudcompute-oct2025/hw3-testfile.txt
```
- `// verify the file had been uploaded into s3 bucket`

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % aws s3 ls shwetaleegadekar-cloudcompute-oct2025
2025-10-13 14:33:30          13 hw3-testfile.txt
```
- `// delete the bucket`

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % aws s3 rb s3://shwetaleegadekar-cloudcompute-oct2025 --force
delete: s3://shwetaleegadekar-cloudcompute-oct2025/hw3-testfile.txt
```
- `// verify that your bucket is deleted`

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % aws s3 ls
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % vi index.html
```

1.2 answer the question: why do we need to add "-- force" to delete the bucket? (5 pts)

We need to add --force when deleting an S3 bucket because by default, AWS does not allow deleting a bucket that still contains objects. The --force flag tells the AWS CLI to first delete all objects inside the bucket and then delete the bucket itself in one command. Without --force, the delete command would fail if the bucket isn't empty.

2. Document the creation of AWS S3 bucket to host a website that includes: (25 pts)

2.1 the source code of index.html file and error.html file. (5 pts)

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % cat index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Hello SCU – Assignment 3</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      background-color: #f5f5f5;
      margin-top: 100px;
    }
    img {
      width: 200px;
      border-radius: 12px;
      margin-top: 20px;
    }
    h1 {
      color: #c80000;
    }
  </style>
</head>
<body>
  <h1>Hello SCU! 🎓</h1>
  <p>Welcome to my AWS S3 Static Website – Assignment 3</p>
  
</body>
</html>
```

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % █
```

2.2 what is the ARN of your S3 bucket? (5 pts)

arn:aws:s3:::shwetaleegadekar-cloudcompute-oct2025

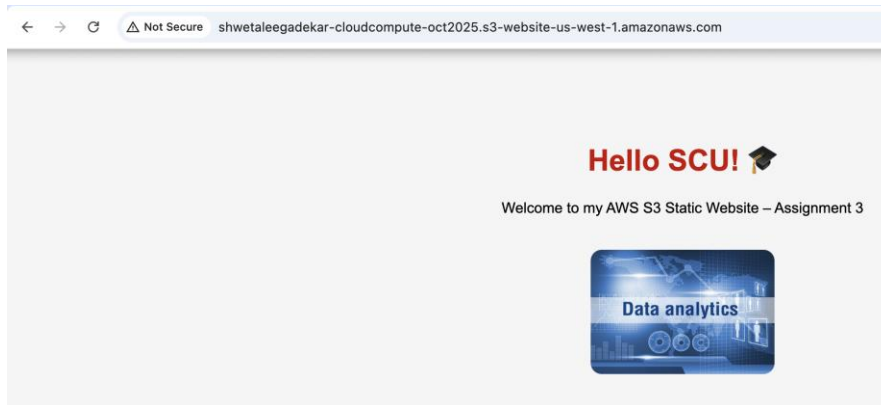
2.3 what is the webhost url? (5 pts)

<http://shwetaleegadekar-cloudcompute-oct2025.s3-website-us-west-1.amazonaws.com>

2.4 copy and paste the content (json format) of the bucket policy you created through AWS Policy Generator (5 pts)

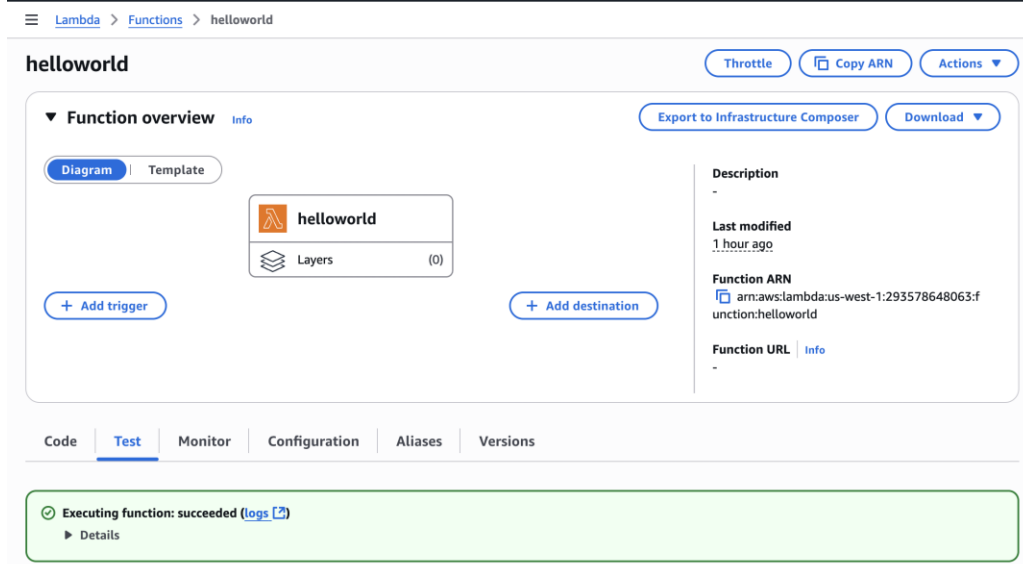
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::shwetaleegadekar-cloudcompute-oct2025/*"
    }
  ]
}
```

2.5 the snapshot of the browser with the webhost url (5 pts)



3. Document the creation of helloworld lambda function and invoke it using aws CLI that includes: (30 pts)

3.1 Screenshot of the helloworld lambda function page with its ARN shown (10 pts)



3.2 Screenshot of the running 'aws lambda invoke ...' command from local terminal of step 8.6 (10 pts)

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % aws lambda invoke --function-name arn:aws:lambda:us-west-1:293578648063:function:helloworld response.json
{
  "statusCode": 200,
  "executedVersion": "$LATEST"
}
```

3.3 Screenshot of the response.json file content generated by 'aws lambda invoke...' command (5 pts)

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % cat response.json
{"statusCode": 200, "body": "\"Hello from Lambda!\""}
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % aws s3 ls
```

3.4 Screenshot of the Cloudwatch console show the /aws/lambda/helloworld Log streams (5 pts)

/aws/lambda/helloworld

Actions

View in Logs Insights

Start tailing

Search log group

▼ Log group details

Log class

Info

Standard

ARN

arn:aws:logs:us-west-1:293578648063:log-group:/aws/lambda/helloworld:*

Creation time

2 hours ago

Retention

Never expire

Stored bytes

-

Metric filters

0

Subscription filters

0

Contributor Insights rules

-

KMS key ID

-

Anomaly detection

Configure

Data protection

-

Sensitive data count

-

Custom field indexes

Configure

Transformer

Configure

<

Log streams

Tags

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Data protection

>

Log streams (2)

Filter log streams or try prefix search

Exact match

Show expired

Info

<

1

>

Log stream

Last event time

2025/10/14/[LATEST]bf237d25d6fc40c9825d2089f8f159a1

2025-10-14 01:46:33 (UTC)

2025/10/14/[LATEST]efb6c1ff397e411f9a2e97aea805fb75

2025-10-14 00:06:32 (UTC)

Log events

Filter events - press enter to search

Clear

1m

30m

1h

12h

Custom

UTC timezone

Display

▶

Timestamp

Message

No older events at this moment. [Retry](#)

▶

2025-10-14T01:36:48.995Z

INIT_START Runtime Version: python:3.13.v63 Runtime Version ARN: arn:aws:lambda:us-west-1::runtime:6e38c6392184a4...

▶

2025-10-14T01:36:49.105Z

START RequestId: 9c274d26-e97b-4801-9f7a-6b9246d2edf2 Version: \$LATEST

▶

2025-10-14T01:36:49.117Z

END RequestId: 9c274d26-e97b-4801-9f7a-6b9246d2edf2

▶

2025-10-14T01:36:49.117Z

REPORT RequestId: 9c274d26-e97b-4801-9f7a-6b9246d2edf2 Duration: 1.77 ms Billed Duration: 109 ms Memory Size: 128...

▶

2025-10-14T01:42:16.737Z

START RequestId: 136fcfff-fc5b-4b9c-b4b6-ca46e356dbee Version: \$LATEST

▶

2025-10-14T01:42:16.779Z

END RequestId: 136fcfff-fc5b-4b9c-b4b6-ca46e356dbee

▶

2025-10-14T01:42:16.779Z

REPORT RequestId: 136fcfff-fc5b-4b9c-b4b6-ca46e356dbee Duration: 40.36 ms Billed Duration: 41 ms Memory Size: 128...

▶

2025-10-14T01:46:05.954Z

START RequestId: 63185fd1-cef5-4a8a-a1bc-6e7a3352595b Version: \$LATEST

▶

2025-10-14T01:46:05.958Z

END RequestId: 63185fd1-cef5-4a8a-a1bc-6e7a3352595b

▶

2025-10-14T01:46:05.958Z

REPORT RequestId: 63185fd1-cef5-4a8a-a1bc-6e7a3352595b Duration: 3.17 ms Billed Duration: 4 ms Memory Size: 128 M...

▶

2025-10-14T01:46:33.808Z

START RequestId: 6ad4a156-d440-405f-a040-3360b31578dc Version: \$LATEST

▶

2025-10-14T01:46:33.818Z

END RequestId: 6ad4a156-d440-405f-a040-3360b31578dc

▶

2025-10-14T01:46:33.818Z

REPORT RequestId: 6ad4a156-d440-405f-a040-3360b31578dc Duration: 8.71 ms Billed Duration: 9 ms Memory Size: 128 M...

No newer events at this moment. [Auto retry paused.](#) [Resume](#)

4. Document the creation of a listen_to_s3_new_object lambda function that will be triggered by s3 object creation that includes: (30 pts)

4.1 Screenshot of the listen_to_s3_new_object lambda function page with its ARN and s3 event trigger ARN shown (10 pts)

listen_to_s3_new_object Throttle Copy ARN Actions

Function overview Info

Diagram Template

Layers (0)

S3 + Add trigger

+ Add destination

Description

Last modified
5 minutes ago

Function ARN
arn:aws:lambda:us-west-1:293578648063:function:listen_to_s3_new_object

Function URL Info

4.2 Screenshot running the aws CLI to create a s3 object and followed immediately with **\$ date** command to record the current date/time. (For Windows host, use “echo %date%- %time%” to capture the time) (10 pts)

```
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % aws s3 cp DataAnalytics.png s3://shwetaleegadekar-cloudcompute-oct2025
upload: ./DataAnalytics.png to s3://shwetaleegadekar-cloudcompute-oct2025/DataAnalytics.png
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ % date
Mon Oct 13 20:19:39 PDT 2025
(base) shwetaleegadekar@Shwetalees-MacBook-Air-3143 ~ %
```

4.3 Screenshot of the CloudWatch logs of listen_to_s3_new_object lambda function triggered by the s3 object creation (10 pts)

- Note: Make sure you expand the CloudWatch log so it shows the details ...
 - In the log file, highlight the date/time (should be very close to the date/time reported by previous step)
 - In the log file, highlight the file name that being copy to the s3 bucket.

> /aws/lambda/listen_to_s3_new_object > 2025/10/14/[\$LATEST]55caab48a1014980bfcfab6a832f9774

Log events Actions Start tailing Create metric filter

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Clear 1m 30m 1h 12h Custom UTC timezone Display

Timestamp	Message
	No older events at this moment. Retry
2025-10-14T03:13:37.724Z	INIT_START Runtime Version: python:3.13.v63 Runtime Version ARN: arn:aws:lambda:us-west-1::runtime:6e38c6392184a4...
2025-10-14T03:13:38.219Z	START RequestId: fd57bde2-7a49-4bd1-9db8-bb3d4827aa26 Version: \$LATEST
2025-10-14T03:13:38.219Z	[INFO] 2025-10-14T03:13:38.219Z fd57bde2-7a49-4bd1-9db8-bb3d4827aa26 File DataAnalytics.png uploaded to alantztan...
2025-10-14T03:13:38.229Z	END RequestId: fd57bde2-7a49-4bd1-9db8-bb3d4827aa26
2025-10-14T03:13:38.229Z	REPORT RequestId: fd57bde2-7a49-4bd1-9db8-bb3d4827aa26 Duration: 2.05 ms Billed Duration: 494 ms Memory Size: 128...
2025-10-14T03:15:01.072Z	START RequestId: ee38fb2e-b3b2-4bf0-92a3-5376c4d443bf Version: \$LATEST
2025-10-14T03:15:01.073Z	[INFO] 2025-10-14T03:15:01.073Z ee38fb2e-b3b2-4bf0-92a3-5376c4d443bf File DataAnalytics.png uploaded to shwetalee...
2025-10-14T03:15:01.090Z	END RequestId: ee38fb2e-b3b2-4bf0-92a3-5376c4d443bf
2025-10-14T03:15:01.090Z	REPORT RequestId: ee38fb2e-b3b2-4bf0-92a3-5376c4d443bf Duration: 16.90 ms Billed Duration: 17 ms Memory Size: 128...
	No newer events at this moment. Auto retry paused. Resume